The Shaded Region Bistro

Assessment Menu · choose your meal and show what you know

Name: _____

Date: _____

How to order

1. Order **4 Appetizers** (5 points each), **4 Entrées** (10 points each) and **2 Chef's Specials** (20 points each). The most you can earn is **100 points**.
2. Your 10 dishes must include **at least one of each skill**: BOUNDARY, SHADE, WRITE and MODEL (the label next to each dish).
3. Mark your choices on the order form below **and** on the dishes themselves ("My pick"). Only marked dishes are graded, and you cannot swap a dish after turning it in.
4. Show your work in the space for each dish. Points are given for correct steps as well as correct answers.
5. Appetizers and Entrées are on pages 2 and 3. The Chef's Specials are on page 4. Look at every page before you choose!

Appetizers

5 pts each · choose 4

- ☐ **A1** (BOUNDARY)
- ☐ **A2** (SHADE)
- ☐ **A3** (BOUNDARY)
- ☐ **A4** (SHADE)
- ☐ **A5** (WRITE)
- ☐ **A6** (MODEL)

Entrées

10 pts each · choose 4

- ☐ **E1** (BOUNDARY)
- ☐ **E2** (SHADE)
- ☐ **E3** (WRITE)
- ☐ **E4** (SHADE)
- ☐ **E5** (MODEL)
- ☐ **E6** (BOUNDARY)

Chef's Specials

20 pts each · choose 2

- ☐ **S1** (WRITE)
- ☐ **S2** (WRITE)
- ☐ **S3** (SHADE)
- ☐ **S4** (MODEL)

My skill check (circle each skill you have covered): BOUNDARY · SHADE · WRITE · MODEL

Course	Dishes	Points each	Possible	Score (teacher)
Appetizers	4	5	20	
Entrées	4	10	40	
Chef's Specials	2	20	40	
Total			**100**	

Appetizers

5 points each · choose 4 · Warm-ups

Name: _____

A1 BOUNDARY ☐ My pick · 5 pts

Should the boundary line be solid or dashed?
(a) $y \geq x - 3$ **(b)** $y < 2x + 1$

A2 SHADE ☐ My pick · 5 pts

Test the point (0, 0) in $y < 2x + 3$. Is it a solution? Should you shade the side of the line that contains the origin?

A3 BOUNDARY ☐ My pick · 5 pts

The boundary of $y \leq 2x - 1$ is the line $y = 2x - 1$. Find y when x = 0 and when x = 2, and say whether the line is solid or dashed.

A4 SHADE ☐ My pick · 5 pts

Which of the points (2, 3), (1, 1), (0, 0) is a solution of $y > 3x - 2$? Show the test for each.

A5 WRITE ☐ My pick · 5 pts

A **dashed** line passes through (0, −2) and has slope 3. The region **above** the line is shaded. Write the inequality.

A6 MODEL ☐ My pick · 5 pts

Maya has at most $30 for notebooks ($3 each) and pens ($2 each). Write an inequality for n notebooks and p pens. Is (6, 6) a solution?

Entrées

10 points each · choose 4 · Multi-step problems

Name: _____

E1 BOUNDARY ☐ My pick · 10 pts

The boundary of $2x - 3y > 6$ is the line $2x - 3y = 6$.

(a) Find its x- and y-intercepts.

(b) Use them to find its slope.

(c) Is the line solid or dashed? Why?

E2 SHADE ☐ My pick · 10 pts

For $y \le -\frac{1}{2}x + 2$, test the points (0, 0), (4, 2), and (−2, 3). Which are solutions? Then say whether you shade above or below the solid line.

E3 WRITE ☐ My pick · 10 pts

A **solid** line passes through (−2, 1) and (2, 5). The region **below** the line is shaded. Write the inequality in slope-intercept form.

E4 SHADE ☐ My pick · 10 pts

Dana graphed $x - 2y < 4$:

$-2y < -x + 4$

$y < \frac{1}{2}x - 2$, so she shaded **below** the dashed line.

Find her mistake, fix it, and confirm with a test point.

E5 MODEL ☐ My pick · 10 pts

A club sells popcorn for $3 and drinks for $2. It needs to raise **at least $120**. Write an inequality for p popcorns and d drinks. Give two combinations that work and one that does not.

E6 BOUNDARY ☐ My pick · 10 pts

Solve $-2y > 6$. Describe the boundary line (type, equation, solid or dashed) and the shading.

Chef’s Specials

20 points each · choose 2 · Multi-part challenges

Name: _____

S1 WRITE ☐ My pick · 20 pts

A **dashed** line passes through (0, 5) and (4, 2). The origin lies in the shaded region.

(a) Write the inequality in slope-intercept form.

(b) Rewrite it with whole-number coefficients (no fractions).

(c) Is (4, 2) a solution? Is (4, 1)?

S2 WRITE ☐ My pick · 20 pts

A system’s solution region lies **on or below** a solid line through (0, 6) and (3, 0), and **above** a dashed line through (0, −2) and (2, 0).

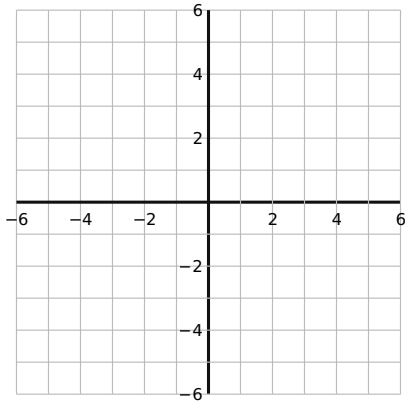
(a) Write the system of inequalities.

(b) Is (1, 1) in the solution region? **(c)** Is (2, 0)?

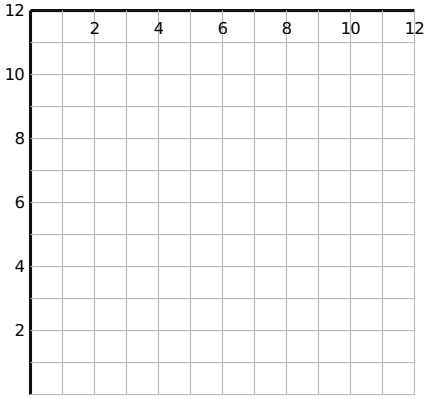
S3 SHADE ☐ My pick · 20 pts

Graph the system on the grid: $y > 2x - 3$ and $x + y \le 6$.

Shade only the region that satisfies both, and label the point where the boundaries cross. Is that point a solution? Explain.



S4 MODEL ☐ My pick · 20 pts

Alex works at most 10 hours a week tutoring (*t* hours, $12 an hour) and dog-walking (*d* hours, $8 an hour). He wants **at least $96**.

(a) Write a system (include $t \ge 0$, $d \ge 0$). **(b)** Graph and shade it (*t* across, *d* up). **(c)** Do 5 tutoring and 4 dog-walking hours work?